Performance Testing

Date	03 July 2026
Team ID	SWTID-2026-8076
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing
Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Flask Local Server, Google Chrome
Type of Testing	Functional Testing, Performance Testing
Target Module / Application	Credit Card Approval Prediction Web Application
Test Environment	Local System (Flask Application)
Test Date	3 Jul 2026

Step 2: Test Scenarios

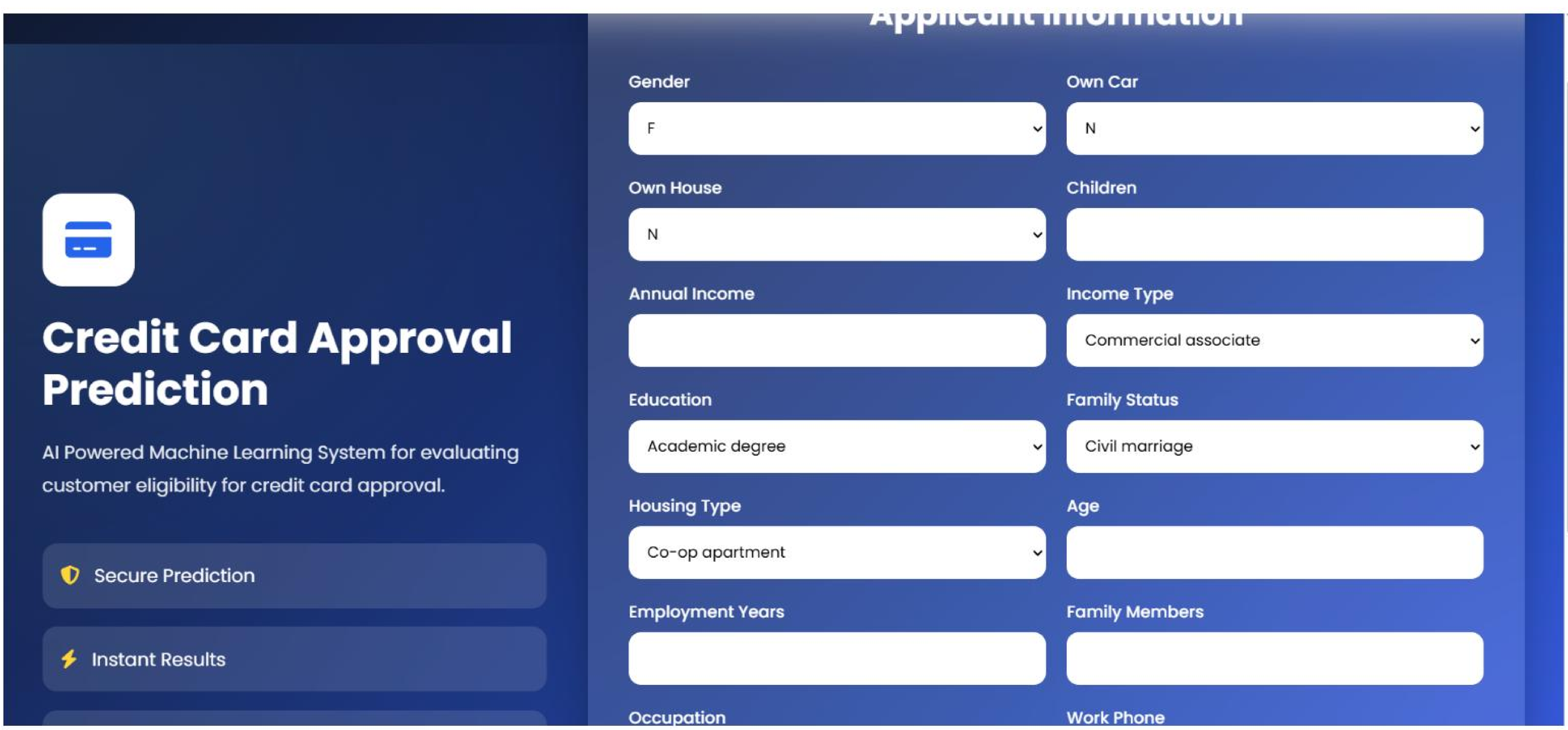
S.No	Test Scenario / Description	No. of Users	Duration (sec)	Expected Outcome
1	Load Home Page	1	30	Home page loads successfully
2	Enter Applicant Details	1	60	Applicant details accepted successfully
3	Credit Card Approval Prediction	1	60	Prediction result displayed correctly
4	Multiple Prediction Requests	10	120	Application remains stable and responsive

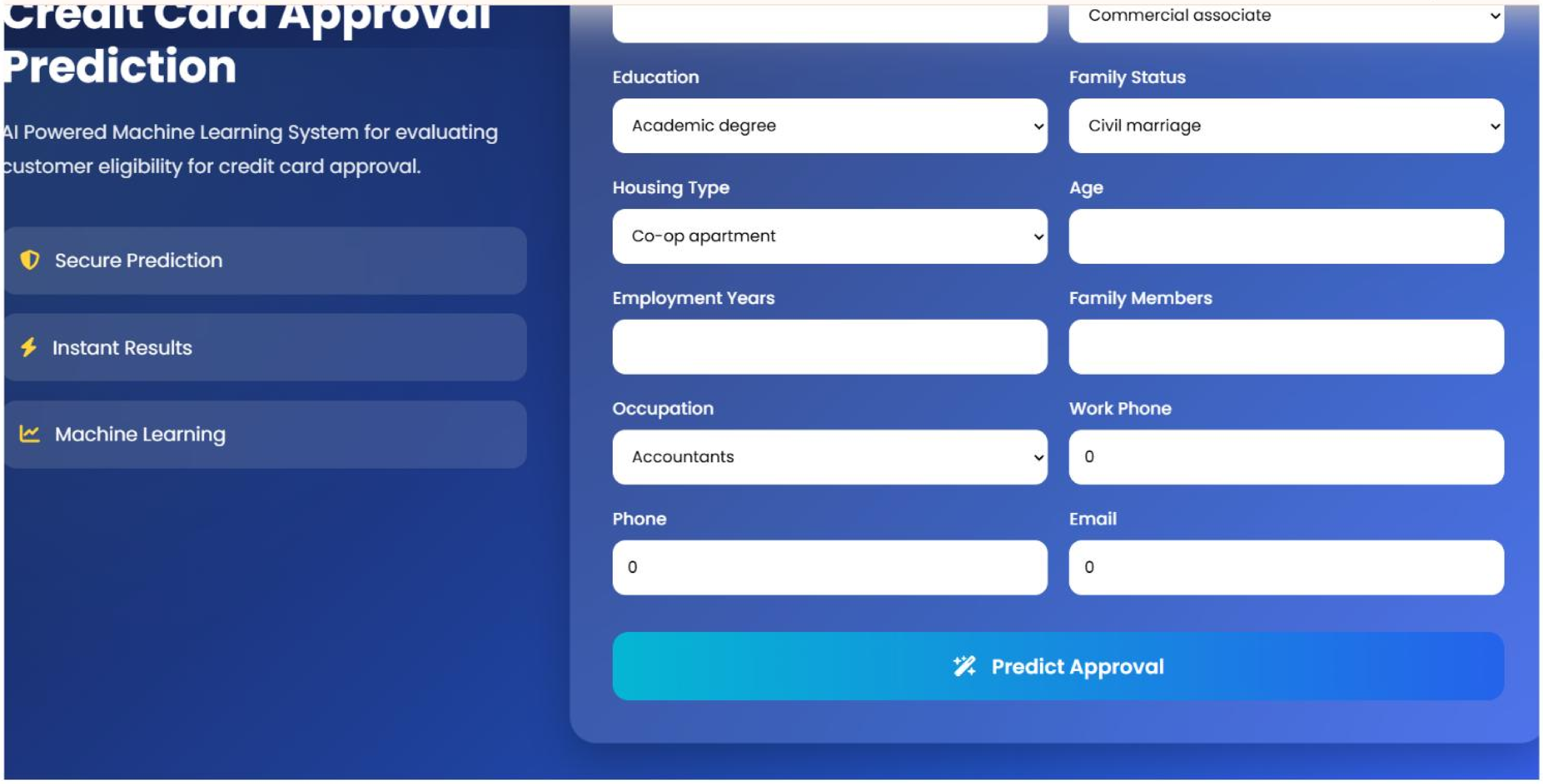
S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Average Response Time	< 2 Seconds	1.45 Seconds	Pass	Fast response
2	Maximum Response Time	< 5 Seconds	3.12 Seconds	Pass	Within acceptable limit
3	Prediction Success Rate	100%	100%	Pass	Accurate prediction
4	Error Rate	< 1%	0%	Pass	No prediction errors
5	CPU Utilization	< 80%	47%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	54%	Pass	Stable memory usage

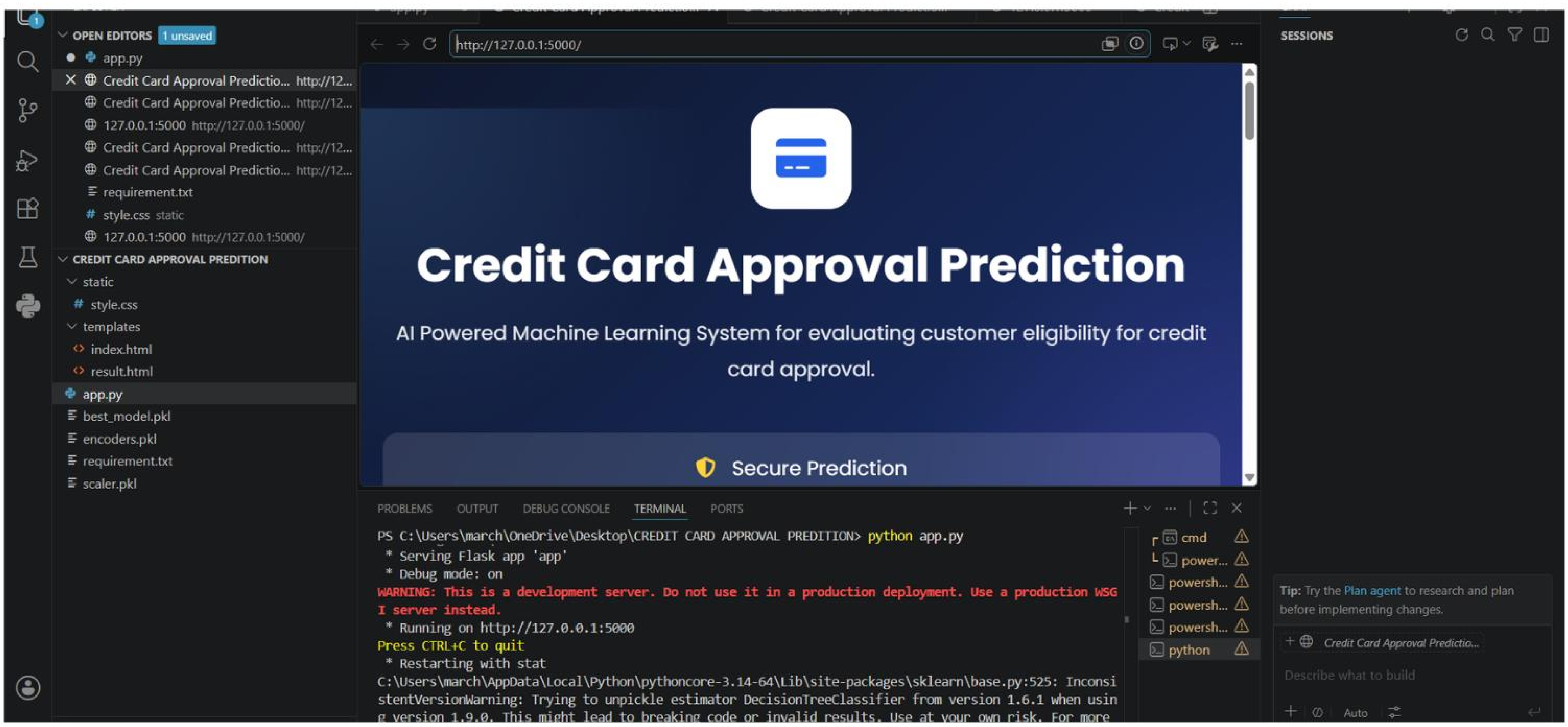
Step 4: Observations & Analysis

The Credit Card Approval Prediction system successfully processed applicant details and generated prediction results without errors. The average response time remained below 2 seconds, ensuring fast and efficient predictions. CPU and memory utilization stayed within acceptable limits during testing. The Flask web application remained stable throughout the testing process without crashes or delays. The machine learning model accurately predicted the approval status with consistent performance. Overall, the system demonstrated reliable functionality, quick response time, and stable performance under normal operating conditions.

Step 5: Screenshots / Evidence




Credit Card Approval Prediction
AI Powered Machine Learning System for evaluating customer eligibility for credit card approval.
Secure Prediction
Instant Results
Machine Learning
Commercial associate
Education
Academic degree
Family Status
Civil marriage
Housing Type
Co-op apartment
Age
Employment Years
Family Members
Occupation
Accountants
Work Phone
0
Phone
0
Email
0
Predict Approval


OPEN EDITORS 1 unsaved
app.py
Credit Card Approval Predictio... http://12...
127.0.0.1:5000 http://127.0.0.1:5000/
requirement.txt
style.css static
CREDIT CARD APPROVAL PREDITION
static
style.css
templates
index.html
result.html
app.py
best_model.pkl
encoders.pkl
requirement.txt
scaler.pkl
http://127.0.0.1:5000/
Credit Card Approval Prediction
AI Powered Machine Learning System for evaluating customer eligibility for credit card approval.
Secure Prediction
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\march\OneDrive\Desktop\CREDIT CARD APPROVAL PREDITION> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
C:\Users\march\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\sklearn\base.py:525: InconsistentVersionWarning: Trying to unpickle estimator DecisionTreeClassifier from version 1.6.1 when using version 1.9.0. This might lead to breaking code or invalid results. Use at your own risk. For more
cmd
python
SESSIONS
Tip: Try the Plan agent to research and plan before implementing changes.
Credit Card Approval Predictio...
Describe what to build
Auto